AVIATION PHOTOGRAPHY TIPS GUIDE

Camera settings • Shooting modes • Aircraft tracking • Composition • Editing

A practical field guide for photographing airliners, general aviation, helicopters, military aircraft, and static displays.

Core idea: choose settings for the aircraft's movement and the light, then keep your attention on tracking the aircraft smoothly and composing the shot.

QUICK START

- **Aircraft in flight:** start in Shutter Priority (S/Tv) around 1/1000 s for jets, with AF-C/AI Servo and burst shooting.
- **Propeller aircraft:** try 1/250–1/500 s to show some propeller motion; use slower speeds for more blur if you can track smoothly.
- **Static aircraft:** use Aperture Priority (A/Av) around f/5.6–f/8, or Manual mode on a tripod when the light is steady.
- **Changing light:** Shutter Priority with Auto ISO is a simple, flexible starting point.

Settings are starting points, not rules. Adjust for aircraft speed, focal length, light, and the look you want.

1. General Camera Settings and Fundamentals

Aviation photography combines moving subjects, long lenses, and changing light. Use these settings as a starting point:

- **Autofocus:** Continuous AF (AF-C on Nikon/Sony/Fujifilm, AI Servo on Canon) keeps updating focus while the aircraft moves.
- **Drive mode:** High-speed continuous burst for short sequences during takeoff, landing, turns, and other key moments.
- **Aperture:** f/5.6–f/8 is a useful range for many telephoto lenses. Open wider in low light; stop down for more depth of field on static aircraft.
- **File format:** RAW gives you more flexibility with exposure, white balance, and noise. RAW + JPEG is useful if you also want quick shareable files.
- **Metering:** Evaluative/matrix metering works well in most situations. Check exposure compensation when bright skies or dark aircraft fool the meter.
- **White balance:** Auto is a good starting point when shooting RAW; fine-tune later.
- **Stabilization:** Use lens/body stabilization when helpful for handheld shots. Follow the lens maker's advice for panning modes.

Exposure triangle in brief

Shutter speed controls motion blur, **aperture** affects light and depth of field, and **ISO** affects sensitivity and image noise. For aircraft in flight, prioritize a shutter speed fast enough to keep the aircraft sharp, then adjust aperture and ISO for exposure.

2. Which Shooting Mode Should You Use?

For aviation, start with **Shutter Priority** for aircraft in flight and **Aperture Priority** for aircraft on the ground. The letters vary slightly by camera brand.

Mode	What you control	Best use
Shutter Priority (S/Tv)	You choose shutter speed; camera chooses aperture. ISO can be automatic or manual.	Aircraft in flight, panning, propeller blur, changing light.
Aperture Priority (A/Av)	You choose aperture; camera chooses shutter speed. ISO can be automatic or manual.	Static displays, detail shots, and depth-of-field control.
Manual (M)	You choose shutter speed and aperture; ISO may be manual or Auto ISO.	Steady light, tripod work, night scenes, consistent exposure.
Program (P)	Camera chooses a shutter/aperture combination; you can often shift it.	Quick snapshots when you need a simple setup.

Recommended default: Shutter Priority + Auto ISO + Continuous AF + high-speed burst

How to set up Shutter Priority for aircraft in flight

- **1. Turn the mode dial** to S (many Nikon, Sony, and Fujifilm cameras) or Tv (Canon). Some cameras let you select it in a menu.
- **2. Choose a shutter speed.** Start at 1/1000 s for airliners. Try 1/1600–1/2000 s for fast jets or rapid action. Around 1/500 s can work for slower aircraft in good light.
- **3. Set ISO to Auto** if clouds and sunlight are changing. If your camera supports an Auto ISO limit, choose a maximum that keeps noise acceptable for your camera.
- **4. Select Continuous AF** (AF-C/AI Servo). Use a tracking or zone focus area you can keep over the aircraft; test subject detection before relying on it.
- **5. Select continuous/burst drive** and take short bursts at important moments instead of holding the shutter for every pass.
- **6. Take a test shot** and zoom in to check sharpness. Check the histogram/highlights if available. Adjust exposure compensation or ISO settings if the exposure is off.

How to set up Aperture Priority for static aircraft

- **1. Turn the dial** to A or Av.

- **2. Choose an aperture.** Start at f/5.6–f/8 for an individual aircraft. Try f/8–f/11 when you need more of the scene in focus. Very small apertures such as f/16 may soften fine detail on some cameras.
- **3. Use Auto ISO** when outdoor light changes, or a low ISO in bright conditions when practical.
- **4. Check the shutter speed** selected by the camera. If it is too slow for handheld shooting, raise ISO or open the aperture.
- **5. Focus on a key detail** such as the cockpit or nearest engine and check the depth of field.

How to use Manual mode

- Use Manual when light is steady or you need the same exposure from frame to frame, such as a static display or night scene.
- Set shutter speed for motion, choose aperture for depth of field, then adjust ISO until exposure looks right.
- **Manual + Auto ISO** is a useful compromise: you lock shutter speed and aperture while the camera adapts ISO to changing light. Exposure compensation behavior varies by camera.
- If light changes quickly during an aircraft pass, Shutter Priority may be easier than fully manual exposure.

Exposure compensation: a quick brightness adjustment

In an automatic exposure mode, use the +/- control to ask the camera to make the image brighter or darker. Bright white aircraft against dark backgrounds and dark aircraft against bright skies can confuse metering. Review the image and histogram instead of assuming one compensation value always works.

Where to find the controls

The mode dial usually shows P, S/Tv, A/Av, and M. AF mode, drive mode, ISO, and Auto ISO are commonly on dedicated buttons, quick menus, or the main menu. If a control is hard to find, search your camera manual for “exposure mode,” “AF-C/AI Servo,” “drive mode,” or “Auto ISO.”

Mode quick guide

Situation	Mode	Starting point
Airliner in flight	S/Tv + Auto ISO	1/1000 s, AF-C/AI Servo, burst
Fast jet / aerobatics	S/Tv + Auto ISO	1/1600–1/2000 s
Propeller aircraft	S/Tv + Auto ISO	1/250–1/500 s for visible prop motion
Panning a landing aircraft	S/Tv	1/125–1/320 s; smooth tracking
Static aircraft outdoors	A/Av	f/5.6–f/8; Auto ISO as needed
Cockpit / detail shots	A/Av	f/4–f/8 depending on depth of field
Tripod / night scene	M	Set shutter and aperture; adjust ISO

3. Shutter Speeds by Aircraft Type

Fast shutter speeds help freeze motion; slower speeds can show motion, especially in propellers and panning shots.

Subject / effect	Starting shutter speed	Notes
Airliners in flight	1/800–1/1600 s	Try 1/1000 s first; increase for fast passes or long focal lengths.
Fast jets / aerobatics	1/1600–1/3200 s	Use fast speeds for sharp action and abrupt direction changes.
General aviation	1/500–1/1000 s	Slower speeds may work with smooth tracking and good light.
Turboprops	1/320–1/1000 s	Balance aircraft sharpness against propeller motion.
Helicopters	1/125–1/500 s	Slower speeds often show rotor movement.
Propeller blur	1/125–1/320 s	Results depend on propeller speed and angle.
Panning	1/80–1/320 s	Choose based on aircraft speed and tracking skill.
Static aircraft handheld	1/250 s or faster	Long lenses may need faster speeds.

If unsure, take several frames at different shutter speeds. Inspect the aircraft at high magnification and keep the sharpest result.

4. Aircraft-Specific Photography Techniques

Airliners

- For arrivals and departures, use a fast shutter speed and leave space in front of the aircraft's direction of travel.
- Look for turns, gear retraction, flap movement, and distinctive airport backgrounds.
- Watch for heat haze from pavement, rooftops, and engines; choose a cooler or less obstructed angle if possible.

General aviation and turboprops

- A slightly slower shutter can show propeller motion, but prioritize a sharp fuselage and cockpit.
- Anticipate the aircraft's path and pan smoothly instead of waiting until it is directly in front of you.

Helicopters

- Use slower shutter speeds for natural-looking rotor blur; this takes practice and may require several attempts.
- Track steadily and keep focus on the body or cockpit rather than the moving rotor.

Military aircraft and fast action

- Use fast shutter speed and continuous autofocus; pre-focus or position the focus area when you know where the aircraft will pass.
- Use short bursts around interesting moments and leave extra space for unpredictable turns.

5. Panning for Motion

Panning technique

- Panning follows a moving aircraft with the camera at a relatively slow shutter speed. The aircraft can stay fairly sharp while the background streaks.
- Start around 1/250 s for slower aircraft or 1/320 s for faster aircraft, then experiment with slower speeds as your tracking improves.
- Stand balanced and rotate smoothly from your hips. Follow the aircraft before pressing the shutter and continue moving after the burst.
- Keep the camera moving at the aircraft's apparent speed. Avoid jerking when pressing the shutter.
- Use Continuous AF and a short burst. Expect fewer sharp frames than with fast shutter speeds.

6. Autofocus and Burst Shooting

Practical focus tips

- AF-C/AI Servo updates focus as the aircraft moves.
- Use a zone or tracking area that leaves room to keep the aircraft inside the active area.
- If your camera offers aircraft/vehicle detection, test it in advance; detection helps but does not replace checking focus.
- High-speed burst is useful for brief moments. Short bursts save card space and make reviewing easier.
- Acquire focus early, keep the aircraft in the focus area, and avoid foreground objects crossing your view.

7. Composition and Framing

Build a stronger frame

- Leave space in front of the aircraft so it has room to fly into the image.
- Use the rule of thirds when it improves the composition; a centered aircraft can also work for symmetry.
- Watch for poles, buildings, wires, and busy details intersecting the aircraft.
- Capture a tight aircraft portrait, a wider environmental image, and details such as engines, landing gear, or markings.
- Check the frame edges so the nose, tail, and wingtips are not accidentally clipped.
- Front or side lighting often reveals shape and detail; backlighting can work creatively but may need exposure adjustment.

8. Airport Spotting and Scouting Locations

Before you go

- Check flight-tracking services such as Flightradar24 or FlightAware to anticipate traffic; not every flight will necessarily appear.
- Use spotting guides and airport information to learn public viewing areas and likely runway directions.
- Check weather, sun angle, cloud cover, and wind. Runway use can change with wind, procedures, and traffic.
- Use a lens hood and avoid hot pavement or exhaust where possible to reduce flare and heat haze.
- Respect airport boundaries and posted restrictions. Never enter restricted areas or aircraft movement areas.

9. Night and Low-Light Photography

Low-light tips

- Use a wider aperture such as f/2.8–f/5.6 if available, and raise ISO as needed to maintain a suitable shutter speed.
- For moving aircraft, avoid a shutter speed so slow that the aircraft becomes blurred. A noisy sharp image is often more useful than a clean but motion-blurred one.
- For static aircraft, a tripod and Manual mode can help. Use a remote release or timer if camera shake is a problem.
- Review bright lights carefully so runway and aircraft lights do not become featureless patches.
- During long exposures, moving aircraft will blur.

10. Static Displays and Detail Shots

Ground photography

- Use Aperture Priority around f/5.6–f/8 for a single aircraft; stop down if more depth of field is needed.
- Focus on details that tell a story: cockpit windows, engine fan blades, landing gear, rivets, and markings.
- Look for clean backgrounds and angles that separate the aircraft from crowds, barriers, and signs.
- Inside cockpits or cabins, follow event rules and be respectful of other visitors.

11. Air-to-Air Photography

Safety first

- Air-to-air photography requires professional planning, approved operations, and strict aviation safety procedures. Do not attempt it without qualified pilots, appropriate permissions, and a safe coordinated plan.
- Secure the camera strap and keep equipment controlled.
- Follow the pilot's instructions. Never distract the pilot or interfere with aircraft controls.

- Choose settings before the shoot when possible; do not handle gear in a way that creates a safety risk.

12. Equipment Recommendations

Useful gear

- **Telephoto lens:** roughly 100–400 mm or 150–600 mm is useful for spotting, depending on sensor size and distance.
- **Standard zoom:** a 24–70 mm or similar lens works for static displays, hangars, and wider scenes.
- **Memory cards:** use reliable cards with enough capacity and speed for RAW burst shooting.
- **Spare batteries:** cold weather, long sessions, and frequent image review can drain batteries faster.
- **Lens hood and cleaning cloth:** help manage flare and keep glass clear.
- **Comfortable strap or bag:** useful for long sessions; keep equipment secure and out of walkways.

13. Common Problems and Solutions

Problem	Possible cause	What to try
Aircraft looks soft	Shutter too slow, focus missed, heat haze, or camera shake.	Raise shutter speed, use AF-C/AI Servo, track smoothly, or change position.
Image too dark	Metering reacted to bright sky or compensation is too low.	Check histogram, adjust compensation, or raise ISO.
Image too bright	Bright aircraft or strong sunlight confused metering.	Lower compensation and check highlights.
Propeller looks frozen	Shutter is too fast for visible prop motion.	Try slower shutter and pan smoothly.
Too much noise	ISO is high or image is underexposed.	Open aperture, choose suitable shutter, and expose accurately.
Busy background	Spotting angle or framing.	Change position, use a longer lens, or wait for a cleaner background.
Heat haze distortion	Hot air between camera and aircraft.	Photograph earlier, move closer only where legal and safe, or change angle.

14. Editing Aviation Photographs

Editing workflow

- Cull first: remove accidental shots and keep frames with the best focus and timing.
- Adjust exposure, highlights, shadows, white balance, and contrast before heavy sharpening.
- Use lens corrections and crop for composition, but avoid excessive cropping that reveals noise or softness.
- Apply noise reduction carefully; too much can erase fine aircraft details.
- Popular tools include Adobe Lightroom, Photoshop, Darktable, and RawTherapee.

15. Aviation Photography Safety and Etiquette

At the airport

- Follow airport signs, barriers, local rules, and staff instructions.
- Stay outside restricted and movement areas; never cross a fence or enter an unsafe location for a photo.
- Keep tripods and bags out of walkways and emergency routes.
- Do not use flash in ways that could distract pilots or airport personnel.
- Respect other spotters and event visitors, and ask before photographing identifiable people closely.

- Use extra care near roads and parking areas; never step into traffic while tracking an aircraft.

16. Quick-Reference Settings Cheat Sheet

Scenario	Mode	Starting settings
Airliner in flight	S/Tv + Auto ISO	1/1000 s, AF-C/AI Servo, burst
Fast jet / action	S/Tv + Auto ISO	1/1600–1/2000 s or faster if light allows
Propeller aircraft	S/Tv + Auto ISO	1/250–1/500 s for some prop blur
Helicopter	S/Tv + Auto ISO	1/125–1/500 s for rotor motion
Panning	S/Tv	1/125–1/320 s, Continuous AF
Static aircraft	A/Av	f/5.6–f/8, Auto ISO if needed
Static details	A/Av	f/4–f/8 depending on depth of field
Night static scene	M	Tripod; set shutter and aperture, adjust ISO

If you remember one setup: for aircraft in flight, start with S/Tv at 1/1000 s, Auto ISO, Continuous AF, and high-speed burst. Check the results and adapt to the aircraft and light.

17. Practice Plan and Final Reminder

Practice sessions

- **Session 1:** use a fast shutter speed and focus on sharpness.
- **Session 2:** try propeller blur at several shutter speeds and compare the results.
- **Session 3:** practice panning with a consistent stance and smooth movement.
- **Session 4:** shoot static aircraft with different apertures to see how depth of field changes.
- **Session 5:** review your best and worst images and note shutter speed, aperture, ISO, and autofocus settings.

Final reminder

The goal is not to memorize one perfect setting. Learn how each control affects the photograph, practice consistently, and adapt to the aircraft, light, and conditions.